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Options and Futures

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Fall Term 2025

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Exotic Options

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- Digital (or binary) options
- Path-dependent options

Exotic options

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- A standard (or plain vanilla) option has payoff $(S_T - K)^+$ or $(K - S_T)^+$.
- Other options are termed "exotic" (but are not that exotic in reality!).

Digital (or binary) options

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- Two types of digital options: cash-or-nothing and asset-or-nothing.
- Payoff of a cash-or-nothing option struck at K and expiring at date T:
  – For some pre-determined fixed amount L

$$\text{Payoff of CoND call} = \begin{cases} L & \text{if } S_T \geq K \\ 0 & \text{otherwise} \end{cases}$$

$$\text{Payoff of CoND put} = \begin{cases} L & \text{if } S_T \leq K \\ 0 & \text{otherwise} \end{cases}$$

Using the indicator function, this is $\mathbf{1}_{S_T \geq K} L$ for call, and $\mathbf{1}_{S_T < K} L$ for put.[1]

[1] The indicator function $\mathbf{1}_C$ takes value 1 if the condition C is satisfied, and zero otherwise.

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Pricing a CoND option

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- Pricing CoND options under the BS assumptions is easy, e.g.

$$\begin{aligned}\text{Price of CoND call} &= e^{-r(T-t)}\mathbb{E}^Q\left[\mathbf{1}_{S_T\geq K}L\right]\\ &= e^{-r(T-t)}L\text{Prob}^Q(S_T\geq K)\\ &= e^{-r(T-t)}L\Phi(d_2),\end{aligned}$$

where

$$d_2=\frac{\log(S/K)+(r-\sigma^2/2)(T-t)}{\sigma\sqrt{T-t}}.$$

- $\Phi(d_2)$ is the risk-neutral probability of exercise.
- Hedging such a digital option when it is near the money close to expiration is daunting (at $S=K, \lim_{t\to T}\delta=\infty$).
  - In practice, these positions are typically not hedged.
  - Note that a vertical spread position (call or put spread with strike prices $K+\varepsilon$ and $K-\varepsilon$) comes close to replicating the payoff of a CoND option.

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Asset-or-nothing digital options

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- Payoff of asset-or-nothing digital options:

$$\text{Payoff of AoND call} = \begin{cases} S_T & \text{if } S_T \geq K \\ 0 & \text{otherwise} \end{cases}$$

$$\text{Payoff of AoND put} = \begin{cases} S_T & \text{if } S_T < K \\ 0 & \text{otherwise} \end{cases}$$

- Payoff can be written $\mathbf{1}_{S_T \geq K} S_T$ for call, $\mathbf{1}_{S_T < K} S_T$ for put.

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$\mathbb{E}[\mathbf{1}_{X\geq K}X]$ for X lognormal

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Result:
If X is lognormally distributed, with $\mathrm{Var}(\log X) = \sigma^2(T-t)$, then

$$\mathbb{E}[\mathbf{1}_{X\geq K}X] = \mathbb{E}(X)\Phi(d_1),$$

where

$$d_1 := \frac{\log[\mathbb{E}(X)/K] + \sigma^2(T-t)/2}{\sigma\sqrt{T-t}}.$$

- This is a special case of the following result that we used to derive the BS formula (the proof is in Appendix A):

$$\mathbb{E}[(X-K)^+] = \mathbb{E}[\mathbf{1}_{X\geq K}(X-K)] = \mathbb{E}(X)\Phi(d_1) - K\Phi(d_2),$$

where

$$d_{1,2} := \frac{\log[\mathbb{E}(X)/K] \pm \sigma^2(T-t)/2}{\sigma\sqrt{T-t}}$$

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Pricing AoND options

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- Pricing AoND options in BS is easy, using the above result and the fact (from the FTAP) that $S_t = e^{-r(T-t)}\mathbb{E}_t^Q(S_T)$. For example,

$$\begin{aligned}\text{Price of AoND call} &= e^{-r(T-t)}\mathbb{E}_t^Q[\mathbf{1}_{S_T \geq K} S_T] \\ &= e^{-r(T-t)}\mathbb{E}_t^Q(S_T)\Phi(d_1) \\ &= S_t\Phi(d_1),\end{aligned}$$

where

$$\begin{aligned}d_1 &= \frac{\log[\mathbb{E}_t^Q(S_T)/K] + \sigma^2(T-t)/2}{\sigma\sqrt{T-t}} \\ &= \frac{\log[e^{r(T-t)}S_t/K] + \sigma^2(T-t)/2}{\sigma\sqrt{T-t}} \\ &= \frac{\log(S_t/K) + (r+\sigma^2/2)(T-t)}{\sigma\sqrt{T-t}}.\end{aligned}$$

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Pricing parity for CoND, AoND and standard European options

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- We can replicate a standard option with digital options:

$$\text{standard European call} = \text{AoND call} - \text{CoND call},$$
$$\text{standard European put} = \text{CoND put} - \text{AoND put},$$

where the payoff of the CoND options is $L = K$.

Path-dependent options

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- We will consider a number of options in this class.
  - Barrier options.
  - Asian options.
  - Lookback options.

Barrier options

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- A barrier option is activated (knock-in) or deactivated (knock-out) when the underlying price hits a predefined barrier H during the life of the option.
  – Four versions: "Up-and-in" or "up-and-out" if the trigger H is above initial price S_0; "down-and-in" or "down-and-out" if trigger H below S_0.
- The option being (de)activated is often a standard European call or put (but it can also be an exotic option, e.g. a digital option).
- For example, an up-and-in call option with barrier $H > S_0$, strike K and expiration date T pays some positive amount $(S_T - K)$ under two conditions:
  – $S_t \geq H$ for some $t \leq T$,
  – and $S_T > K$.

Barrier options (cont'd)

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- For a given trigger price, combining a knock-in and a knock-out option gives the payoff of a standard option.
  – No-arbitrage implies, e.g. $C = C_{IN} + C_{OUT}$. This implies that $C_{IN} \leq C$ and $C_{OUT} \leq C$, i.e. barrier options are cheaper.
- Barrier options are very common in structured products. They may seem appealing because of their lower premia. But it doesn't feel good to see a knock-out option disappear!
- An important contractual detail is the frequency at which the underlying price is monitored to check (de)activation.
- Remark: If you are long an up-and-in barrier option and S is just below the barrier H, it is tempting to buy the underlying to push S up and activate the option!
  – In practice, to avoid such price manipulation, what is tracked to assess whether the barrier has been hit or not is not the price itself but some price average over a short time window.

Pricing barrier options

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- Under BS assumptions with continuous monitoring, there are closed-form formulas for European call and put options with a barrier.
  – Need to use the distribution of the maximum/minimum value taken by a Brownian motion over some time interval.

Asian options

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- Two types of Asian options: Asian-price and Asian-strike.
  - Let $\bar{S}$ denote the average level of S between 0 and T.
- Asian-price options: Call payoff $(\bar{S} - K)^+$, put payoff $(K - \bar{S})^+$.
- Worth less than a standard option because the volatility of $\bar{S}$ is lower than the volatility of S_T.
- Buyer's perspective: Using average-price options makes sense if you are exposed to price fluctuations over some time period.
  - Cheaper than hedging with a bunch of options of different maturities.
- Market maker's perspective: Option is very easy to hedge near expiration (very little uncertainty about final payoff).

Asian options (cont'd)

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- Asian-strike options: Call payoff $(S_T - \bar{S})^+$, put payoff $(\bar{S} - S_T)^+$.
- The exact averaging period and mode of averaging (arithmetic vs. geometric) as well as the frequency of observations used to compute the average $\bar{S}$ are important aspects of Asian option contracts.
  – e.g. more frequent averaging makes an Asian-price option less valuable.

Lookback options

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- Define
$$S_{\min} := \min\{S_t : t \in [0, T]\},$$
$$S_{\max} := \max\{S_t : t \in [0, T]\}.$$
- The payoff of a lookback call is $S_T - S_{\min}$.
  – i.e. right to buy at the lowest price realized over the life of the option.
- The payoff on a lookback put is $S_{\max} - S_T$.
  – i.e. right to sell at highest price realized over the life of the option.
- The payoff of a lookback option is higher than that of a standard option $\Rightarrow$ higher option premium.
- Sampling frequency to compute the min/max is again crucial.

General remarks on exotic options

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- Financial engineers are very imaginative $\rightarrow$ endless list of payoffs traded OTC.
- Structurers can tailor payoff profile to meet the specific needs of each client (with hefty margin!).
- For corporate treasurers, exotic options sometimes provide meaningful hedging solutions.

General approach to pricing exotic options

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- FTAP applies: The no-arbitrage price of an exotic option is its expected payoff under the risk-neutral measure, discounted at the riskfree rate.
- Under the lognormality assumption, some closed-form formulas exist.
- More generally, the risk-neutral expectation can be computed by Monte-Carlo simulations, i.e. simulating the risk-neutral process for the underlying asset price(s).
  – Can use the local volatility function $\sigma(S_t, t)$ à la Dupire calibrated on plain vanilla option prices: Under $\mathbb{Q}$,
  $dS_t = rS_t dT + \sigma(S_t, t)S_t dW_t$.

Hedging exotic options

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- Hedging an exotic option as compared with a plain vanilla European option can be easier (e.g. Asian-price option) or harder (e.g. digital, barrier).
- Absent a closed-form formula, delta and the other Greeks have to be computed numerically.
  – e.g. compute option price V for initial price S_0 and for $S_0 + \Delta S$. Delta of the option at S_0 is then approximated by

$$\frac{V(S_0 + \Delta S) - V(S_0)}{\Delta S}.$$

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- Appendix A contains proofs of some results
- This material will not be examined

$\mathbb{E}[(X-K)^+]$ for X lognormal

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- To derive the Black-Scholes formula, we used the following result: Suppose X is lognormally distributed, and $\text{Var}(\log X) = \sigma^2(T-t)$. Then

$$\mathbb{E}[(X-K)^+] = \mathbb{E}(X)\Phi(d_1) - K\Phi(d_2),$$

where Φ is the standard normal cdf and

$$d_1 : \frac{\log[\mathbb{E}(X)/K] + \sigma^2(T-t)/2}{\sigma\sqrt{T-t}}, \quad d_2 := d_1 - \sigma\sqrt{T-t}$$

- To simplify notation, let $\mathbb{E}(\log X) = \alpha$, and $\text{Var}(\log X) = \beta^2$ (i.e. $\beta := \sigma\sqrt{T-t}$)

Proof

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- We can write $\log X = \alpha + \beta z$, where z is standard normal. Thus $X = e^{\alpha+\beta z}$ and $\mathbb{E}(X) = e^{\alpha+\beta^2/2}$. Moreover, $X \geq K$ is equivalent to $\log X \geq \log K$,or

$$\begin{aligned} z &\geq \frac{\log K - \alpha}{\beta} \\ &= \frac{\log K - \log \mathbb{E}(X) + \beta^2/2}{\beta} \\ &= -\frac{\log[\mathbb{E}(X)/K] - \beta^2/2}{\beta} \\ &= -d_2. \end{aligned}$$

- Recall the standard normal density: $\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2}$
- Let Φ be the standard normal cdf.

Proof(cont'd)

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Thus we have:

$$\begin{aligned}
\mathbb{E}[(X-K)^+] &= \int_{-d_2}^{\infty} (e^{\alpha+\beta z} - K)\phi(z)dz \\
&= \int_{-d_2}^{\infty} e^{\alpha+\beta z}\phi(z)dz - K\int_{-d_2}^{\infty}\phi(z)dz \\
&= \int_{-d_2}^{\infty} \mathbb{E}(X)e^{-\frac{1}{2}\beta^2+\beta z}\cdot\frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}z^2}dz - K\int_{-d_2}^{\infty}\phi(z)dz \\
&= \mathbb{E}(X)\int_{-d_2}^{\infty}\frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}(z-\beta)^2}dz - K\int_{-d_2}^{\infty}\phi(z)dz \\
&= \mathbb{E}(X)\int_{-d_2-\beta}^{\infty}\frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}(z-\beta)^2}d(z-\beta) - K\int_{-d_2}^{\infty}\phi(z)dz \\
&= \mathbb{E}(X)\int_{-d_1}^{\infty}\phi(z-\beta)d(z-\beta) - K\int_{-d_2}^{\infty}\phi(z)dz \\
&= \mathbb{E}(X)\Phi(d_1) - K\Phi(d_2).
\end{aligned}$$

Breeden-Litzenberger

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- Let $\varphi(\cdot)$ be the density of S_T under the risk-neutral probability Q The price of a call option at $t = 0$ as a function of the strike price K can be written as:

$$C(K) = e^{-rT}\mathbb{E}^Q[(S_T - K)^+] = e^{-rT}\int_K^\infty (x - K)\varphi(x)dx$$

- Result [Breeden-Litzenberger (1978)]:

$$\varphi(K) = e^{rT}C''(K).$$

Proof of Breeden-Litzenberger result

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Proof: We can write

$$e^{rT}C(K) = \int_K^\infty x\varphi(x)dx - K\int_K^\infty \varphi(x)dx.$$

Differentiating:

$$\begin{aligned} e^{rT}C'(K) &= \frac{d}{dK}\int_K^\infty x\varphi(x)dx - K\frac{d}{dK}\int_K^\infty \varphi(x)dx - \int_K^\infty \varphi \\ &= -K\varphi(K) - K[-\varphi(K)] - \int_K^\infty \varphi(x)dx \\ &= -\int_K^\infty \varphi(x)dx. \end{aligned}$$

Differentiating again:

$$e^{rT}C''(K) = -[-\varphi(K)] = \varphi(K).$$

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- Put-call parity (non-dividend paying stock):

$$C_t = P_t + S_t - e^{-r(T-t)}K$$

- Lognormal property of GBM:
  Suppose $dS_t = \mu S_t dt + \sigma S_t dW_t$. Then the conditional distribution of $\log S_T$ at time t is normal:

$$\log S_T \sim \mathcal{N}\left[\log S_t + \left(\mu - \sigma^2/2\right)(T-t), \sigma^2(T-t)\right]$$

- Black-Scholes formula:
  The price of a European call option at $t < T$ is given by

$$C(S,t) = S\Phi(d_1) - e^{-r(T-t)}K\Phi(d_2)$$

  where $\Phi(\cdot)$ is the standard normal cdf, and

$$d_1 = \frac{\log(S/K) + \left(r + \sigma^2/2\right)(T-t)}{\sigma\sqrt{T-t}}, \quad d_2 = d_1 - \sigma\sqrt{T-t}$$

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- Black-Scholes PDE:
  Consider a derivative on a stock with payoff $g(S_T)$ at time T, whose price at time $t \leq T$ is given by $F(S,t)$. The Black-Scholes PDE is

$$F_t(S,t) + rSF_S(S,t) + \frac{1}{2}\sigma^2 S^2 F_{SS}(S,t) - rF(S,t) = 0$$

- Standard normal density:

$$\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$$

- Expectation of the exponential of a normal random variable:
  If X normally distributed, then

$$\mathbb{E}[\exp(X)] = \exp\left[\mathbb{E}(X) + \frac{1}{2}\operatorname{Var}(X)\right]$$

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